

Project Constitution — Agentic Multimodal RAG

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This document defines the non-negotiable principles of the project. The technical specification and implementation plan must comply with it. If a design decision conflicts with a principle here, the principle wins, or the constitution is amended explicitly.

1. Purpose

Build an agentic Retrieval-Augmented Generation system over the project's PDF corpus. It answers questions with **text, original figures from the PDFs, generated charts (bar, pie, line) and data tables**, with every element traceable to its source page and no loss of information.

2. Core principles

P1 — No information loss

- Every piece of content in a PDF (text, embedded images, vector diagrams, tables and scanned pages) must be captured at ingestion. Nothing is silently dropped.
- Anything intentionally skipped, such as decorative logos, repeated headers or icons, is **logged with a reason**, never discarded silently.
- Figures are stored at native resolution. Vector diagrams are rendered at no less than 200 DPI.

P2 — Provenance for everything

- Every element (text chunk, figure, table, caption, chart data point) carries: **source file, page number, bounding box, element ID**.
- Every block in an answer shows its source. A statement that cannot be cited is not included in the answer.

P3 — Originals, not imitations

- Images shown to the user are the **original figures from the PDFs**, never AI-generated or redrawn versions.
- The system never generates illustrative images.

P4 — Charts must be truthful

- A chart is drawn **only from numbers retrieved from the corpus** or computed deterministically from them. The LLM never invents, estimates or "fills in" values.
- Every chart is accompanied by **its underlying data table and source citation**.
- Each value is flagged **exact** (from text or a table) or **estimated** (read visually off a chart image). Charts containing estimated values are labelled "approximate".
- Chart type must match the data:

- **Pie** only for parts of a whole.
- **Bar** for comparisons across categories.
- **Line** for ordered or temporal series.

P5 — Grounded answers, honest gaps

- The agent answers only from retrieved evidence. If the corpus doesn't contain the answer, it says so plainly.
- The agent must not present general model knowledge as if it came from the documents. If background knowledge is used at all, it is clearly marked as such.

P6 — Models are replaceable

- The VLM, LLM and embedding model are each accessed through a single interface and selected by configuration.
- Changing a model (for example Qwen-VL to another VLM, or one OpenAI model to another) must not require changes to pipeline logic.

P7 — Cost- and compute-conscious

- **GPU is used only for VLM captioning at ingestion**, on a temporary or free GPU (Kaggle, Colab or rented). No GPU at query time.
- Query-time reasoning uses the OpenAI API, which is pay-per-use.
- Expensive work (captioning, embeddings) is **cached by content hash** and never repeated for unchanged content.

P8 — Reproducible, incremental ingestion

- Ingestion is deterministic, and every stage can be re-run on its own.
- Adding a new PDF processes only that PDF.
- Each stage writes inspectable intermediate outputs (JSON/JSONL, PNG) so any step can be audited by hand.

P8a — One source of truth, consistent data

- A single database holds all metadata, captions, tables, vectors and the keyword index. There is no second store that can drift out of sync.
- Each document is written **atomically**: all of its data is committed together, or none of it is. Search never returns results from a half-ingested document, or from a mix of old and new versions.
- The database can always be rebuilt from the source PDFs plus cached intermediate outputs.

P9 — Simplicity first

- Prefer plain Python and direct SDK calls over heavy frameworks, unless a framework removes significant complexity.
- Build the smallest working version end to end first, then improve quality.

P10 — Measured, not assumed

- Quality claims are backed by an evaluation set, scored with RAGAS plus project-specific checks (charts, figures, citations, refusals).
- Changes to prompts, models or retrieval are compared against the previous baseline before being kept.

P11 — Authenticated, accountable access

- Only signed-in users can use the application. Accounts are created by an administrator; there is no public sign-up.
- Passwords are stored only as strong one-way hashes (Argon2id). They are never logged, displayed or stored in plain text.
- A failed sign-in never reveals whether an account exists. Repeated failures lock the account temporarily.
- Every question and every sign-in event is attributable to a user. Each user has daily usage limits that protect the API budget.

P12 — Observable by default

- Every question and every ingestion run is **traced end to end**. Each plan, agent round, tool call, model call, database query and validation step can be inspected afterwards.
- Telemetry follows an open standard (OpenTelemetry), so the viewing tool can be changed without code changes.
- Telemetry never breaks the application and never contains secrets or passwords. It records user content only where explicitly allowed, and is kept only as long as the retention policy says.

3. Working agreements (human ↔ AI assistant)

- **W1 — No code without permission.** The AI assistant does not write or run code until the owner explicitly approves the specific phase or task.
- **W2 — Roadmap before build.** Each phase starts with a short plan. Work begins only after owner approval.
- **W3 — Phase gates.** Each phase ends with a demo or evidence of its acceptance criteria and owner sign-off before the next phase begins.
- **W4 — Transparency.** Failures, skipped steps, cost overruns and known limitations are reported plainly.
- **W5 — Test-driven development, where it fits.** Code is proven by automated tests, not by manual checks.
 - **Deterministic logic** (config, IDs, chunking, search scoring, `compute`, chart validation, the answer validator, citations, sign-in and sessions, limits, database writes) is built **test-first**: a failing test from the acceptance criteria, then code until it passes, then refactoring with all tests green.

- **Exploratory parts** (PDF figure and table detection) are tuned with the owner on the review sheet first. The approved results are then frozen as regression tests.
- **LLM answer quality** is judged by the RAGAS evaluation (P10), not by unit tests.

The full suite runs after every change. Phase summaries report test counts and results.

4. Out of scope (v1)

- Generating new images or illustrations.
- Real-time collaboration.
- Self-service sign-up, email-based password reset, single sign-on and multi-factor authentication. v1 uses admin-created email + password accounts (P11); these may be added later.
- Fine-tuning any model.
- Non-PDF sources (web pages, DOCX, video). These may be added later without violating P6.

5. Amendments

Changes to this document require the owner's explicit approval and are recorded below.

Version	Date	Change
0.1	2026-09-26	Initial draft
0.2	2026-09-26	Added P8a (one source of truth, atomic writes per document), following the owner's decision to use PostgreSQL for all storage
0.3	2026-09-26	Added P11 (authenticated, accountable access) after the owner chose email + password login; out-of-scope list updated
0.4	2026-09-26	Added P12 (observable by default, OpenTelemetry) at the owner's request
0.5	2026-09-26	P10 names RAGAS as the evaluation framework, at the owner's request
0.6	2026-09-27	Added W5 (test-driven development where it fits: TDD for deterministic logic, frozen regression tests for exploratory parsing, RAGAS for LLM quality), at the owner's request